

Supplier Procurement Risk & Demand Analysis Project

Business Requirement Document (BRD)

Prepared By: Karan Badakh

Role: Data Analyst / SCM Analyst

Technology Used: MySQL, Python, Pandas, Power BI

Date: 27 May 2026

1. Project Overview

The organization currently faces challenges in monitoring **supplier performance**, **procurement trends**, **defect rates**, **delivery delays**, and **operational efficiency**. Procurement and **production planning** analysis are **largely manual**, making it difficult for management to **identify supply chain risks** and **take timely business decisions**.

This project **aims** to develop a **centralized Supply Chain Procurement & Supplier Performance Analytics System** using **MySQL, Python, Pandas, NumPy** and **Power BI** to provide **data-driven insights** for **supplier management**, **procurement optimization**, **quality monitoring**, and **production planning**.

2. Business Problem Statement

The current procurement and supplier monitoring process has the following issues:

- No centralized supplier analytics system
- Difficulty identifying high-risk suppliers
- Limited visibility into procurement trends
- Delayed identification of delivery risks
- Lack of supplier quality monitoring
- Manual KPI reporting process
- Inefficient inventory and production planning
- No real-time visibility into supplier defect rates and lead times

These challenges impact:

- production continuity
- inventory availability
- procurement efficiency

- operational cost
- supplier performance management

3. Business Objectives

The project should help the organization to:

- Monitor supplier-wise procurement performance
- Analyse product-wise revenue and order trends
- Identify supplier dependency risks
- Monitor defect rates and quality performance
- Analyse delivery delay risks
- Improve production planning decisions
- Optimize procurement and logistics efficiency
- Track shipping and manufacturing costs
- Improve inventory visibility
- Support data-driven management decisions

4. Project Scope

In Scope

- ✓ Supplier performance analysis
- ✓ Product-wise revenue analysis
- ✓ Order quantity analysis
- ✓ Defect rate monitoring
- ✓ Lead time analysis
- ✓ Shipping efficiency analysis
- ✓ Delivery risk analysis
- ✓ SKU coverage analysis
- ✓ Procurement KPI dashboard
- ✓ SQL stored procedures
- ✓ Power BI dashboard visualization
- ✓ Python-based EDA and data cleaning

Out of Scope

- ✗ Machine learning forecasting
- ✗ Automated supplier notifications

5. Stakeholders

Stakeholder	Responsibility
Production Planning Team	Production scheduling & planning
Purchase Department	Supplier procurement management
Quality Team	Defect monitoring
Logistics Team	Shipping & delivery tracking
Management	Strategic decision-making
Data Analyst	Reporting & dashboard development
IT Team	Database & infrastructure support

6. Data Fields Used

Field	Purpose
Supplier Name	Supplier analysis
Product Type	Product performance analysis
SKU ID	Product coverage analysis
Revenue	Financial KPI
Order Quantity	Procurement analysis
Shipping Cost	Logistics analysis
Manufacturing Cost	Cost analysis
Defect Rate	Quality monitoring
Lead Time	Production planning
Inspection Results	QA analysis
Month	Trend analysis

7. Key Business KPIs

KPI	Business Meaning
Total Revenue	Overall business revenue
Total Order Quantity	Procurement volume
Average Defect Rate	Supplier quality performance
Average Lead Time	Procurement efficiency
Supplier Contribution %	Supplier dependency risk
Shipping Efficiency	Logistics performance
Delivery Delay Risk	Supply chain risk
SKU Coverage	Product availability
Total Profit	Operational profitability
High Defect Supplier Flag	Supplier quality alert

8. Technical Stack

Technology	Purpose
MySQL	Database Connectivity, EDA, analysis & stored procedures
Python	Data cleaning & analysis
Pandas & NumPy	EDA & KPI analysis
Power BI	Dashboard visualization
MYSQL	Data extraction & transformation

9.Existing Problems Solved

Problem	Solution Implemented
Manual procurement analysis	Automated KPI reporting
Supplier performance visibility issue	Supplier-wise dashboards

Problem	Solution Implemented
Difficulty tracking defects	Defect rate analysis
Delivery risk monitoring issue	Delay risk KPI
High supplier dependency	Supplier contribution analysis
Poor logistics visibility	Shipping cost & efficiency KPIs

10. SQL Stored Procedures Developed

The following stored procedures were developed:

- Supplier Performance Analysis
- Supplier Dependency Risk Analysis
- Monthly Procurement Trend Analysis
- Top Supplier Identification
- Demand Fluctuation Analysis
- Supplier KPI Dashboard
- Delivery Delay Risk Analysis
- High Defect Supplier Flag Analysis

11. Dashboard Features

The dashboard includes:

- ✓ KPI Cards
- ✓ Supplier-wise revenue analysis
- ✓ Defect rate monitoring
- ✓ Order Volume Across Suppliers
- ✓ Lead time analysis
- ✓ SKU coverage visualization
- ✓ Defect Rate across supplier
- ✓ Inspection Status across Suppliers

12.Expected Business Benefits

Benefit	Impact
Faster reporting	Reduced manual effort
Better supplier visibility	Improved procurement planning
Reduced supply chain risk	Better operational continuity
Improved quality monitoring	Reduced defect-related losses
Better inventory planning	Reduced stock issues
Data-driven decision-making	Improved management efficiency

13.Conclusion

The Supplier Procurement Risk & Demand Analysis Project provides a centralized analytics solution for monitoring procurement performance, supplier quality, operational efficiency, and supply chain risks. The system enables management to make faster, data-driven decisions for improving production planning, supplier management, and operational performance.